INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Date: 18-11-2024 AN | **Time: 3 Hrs** | **Full Marks: 35**

No. of Students: 46 | **End Autumn Semester Examination**

Sub. No.: CD61002 | **High Performance Scientific Computing**

Instructions: You may carry class lecture notes, slides and notebook, Yousef Saad's book and a print-out/hand-out with syntax and semantics of your preferred programming language and the codes you have developed through the assignments. Use computer terminals for Part B. Part A to be submitted before computers are turned on.

Part A (25 marks) – 110 mins

Q1. i. Write down the biconjugate gradient algorithm. Show the locations in which you have to introduce mpi/openmp/cuda constructs to parallelize it. What type of architecture will be utilized for this (Flynn's taxonomy)?

ii. Discuss the difficulties in parallelization of Tridigaonal matrix algorithm (TDMA)

2+1+2=5

Q2. i. Explain the difference between critical and atomic constructs in OpenMP programming.

ii. Consider two computations- A) Adding two vectors, B) Finding norm of a vector, in which of these two cases you may need to use reduction clause and why?

iii. Why do we need to use private variables to compute a for loop in shared memory program?

2+2+1=5

Q3. i. What is the difference between collective communication and point to point communication in an MPI program? Explain how an MPI_BCAST function call can give better efficiency to multiple MPI_SEND and MPI_RECV calls for the same message?

ii. Discuss the overheads associated with MPI_SEND and MPI_RECV calls?

4+1=5

Q4. i. What is shared memory in a CUDA program and how can it be used to improve performance of the program?

ii. Explain the use of the constructs cudaMalloc in a CUDA program.

iii. A CUDA kernel that runs with large number of threads in a single block is inefficient- why?

2+1+2=5

Q5. i. Why is it not possible to get a linear scalability in parallel computation using a very high number of processors?

ii. Consider a domain decomposition algorithm in a rectangular domain in with 1001 x 251 mesh using 4 processes. Draw the schematic domains with one-cell overlap in each direction. Show the data transfer steps for the 0-th process. What is the volume of data transfer from 0-th process to 1^{st} process. Consider all physical boundaries to be Dirichlet.

2+3=5

Part B (10 marks) – 70 mins

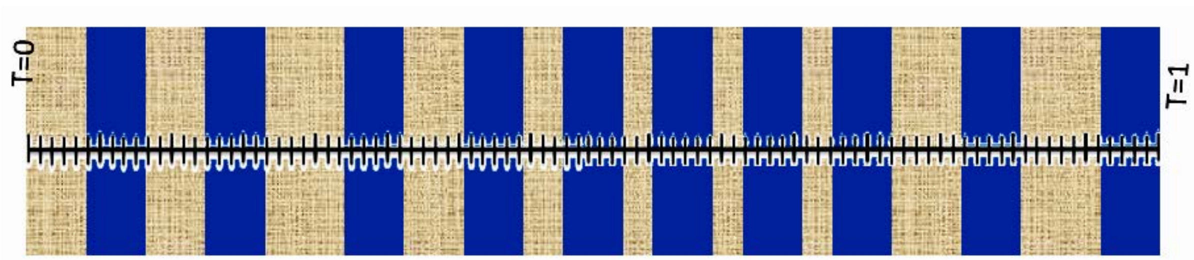
Q6. Consider one-dimensional conduction with heat generation through the composite slab as shown in the figure along with the boundary conditions. The FDM equation for a grid point i is given as:

$T_{i-1} - 2T_i + T_{i+1} = 2dx^2$ if the point i is in the shaded grey zone

$T_{i-1} - 2T_i + T_{i+1} = 0.25dx^2$ if the point i is in the blue zone

$T_{i-1} - 3T_i + 2T_{i+1} = 0$ if the point i is in the interface

Explain how you can solve for temperature at all internal points using CUDA accelerated Jacobi solver. Show the steps involved. Will your algorithm give warp divergence? Then how to make it free from warp divergence? Discuss when will you need to copy data between host and device.



Or

Q7. Consider the matrix $A_{(1025x1025)}$ as $a_{ij} = (-1)^i + \frac{1}{2}(-1)^{2i+j}$. Find and (i) L_2 norm of A and (ii) L_∞ norm of A^TA using 4 processors. Confirm a near equitable load balancing. Report the respective speed-ups using MPI and OpenMP. In which case (i or ii) do you see a better speed up and why? You need to upload the working code in MS Team against Assignment-5.